Synthetic OCR Validation Corpus

This document exists to be converted, not read. Prose paragraphs are present so that crops have neighbouring text to be located by, mirroring how a real article reads around its display elements.

CORPUSMARK-EQ-001 A numbered display equation, the standard weighted arithmetic mean:

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i\, x_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

The weights are normalised so that the result stays bounded.

CORPUSMARK-EQ-002 A fraction combined with an improper integral:

$$\frac{\partial u}{\partial t} = \int_0^\infty e^{-\lambda t}\, d\mu(\lambda) \tag{2}$$

This form appears throughout the transform literature.

CORPUSMARK-EQ-003 Nested subscripts and superscripts:

$$\sigma_{i,j}^2 = \frac{1}{N-1} \sum_{k=1}^{N} \left(x_k^{(i)} - \bar{x}^{(i)} \right)^2 \tag{3}$$

Sample variance, written the long way.

CORPUSMARK-EQ-004 A multi-line aligned derivation, which occupies several rows and is therefore much taller than a single-line equation:

$$\begin{aligned} f(x) &= (x+1)^2 && (4) \\ &= x^2 + 2x + 1 && (5) \\ &= x^2 + 2x + 1 + 0 && (6) \end{aligned}$$

Each step follows from the previous one.

CORPUSMARK-EQ-005 A matrix, which is tall relative to its width:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \tag{7}$$

Matrices are the classic case where a wide-and-short assumption fails.

CORPUSMARK-EQ-006 A piecewise definition:

$$H(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases} \tag{8}$$

The Heaviside step, defined by cases.

CORPUSMARK-EQ-007 Set-theoretic and logical operators, which rely on symbol fonts:

$$\forall \varepsilon > 0\ \exists \delta > 0 : |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon \tag{9}$$

The usual definition of continuity.

CORPUSMARK-EQ-008 Greek letters mixed with relations:

$$\alpha \wedge \beta \subseteq \Gamma \cup \Delta, \qquad \theta \in \Theta \setminus \{\emptyset\} \tag{10}$$

Symbols here span several Unicode blocks.

CORPUSMARK-EQ-009 An unnumbered display equation:

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e$$

Unnumbered forms are common in proofs.

CORPUSMARK-EQ-010 An equation carrying units and upright text:

$$E_{\text{total}} = mc^2 + \tfrac{1}{2}mv^2 \quad [\mathrm{J}] \tag{11}$$

Mixed italic and upright content in one expression.

CORPUSMARK-EQ-011 A display equation with an embedded natural-language phrase, which the model must reproduce as words rather than garble into symbols:

$$g(x) \geq 0 \quad \text{for all } x \in \mathbb{R}, \qquad \text{and } g(0) = 0. \tag{12}$$

Text inside mathematics is a distinct recognition case from surrounding prose.

CORPUSMARK-EQ-012 The next display equation is introduced by a table cross-reference in running prose. The sentence begins with a table label but is not a caption for the equation, and the equation is a wide single-line strip – the exact shape that was being mislabelled as a table.

Table 3 reports the fitted coefficients of the following linear predictor:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon \tag{13}$$

The residual term is assumed to be mean zero.

CORPUSMARK-EQ-013 The next display equation is introduced by a figure cross-reference sentence, the analogous trap for figures rather than tables.

Figure 4 shows that the estimator can be written in the closed form

$$\hat{\theta} = \left(X^\top X\right)^{-1} X^\top y \tag{14}$$

which is the ordinary least squares solution.

CORPUSMARK-EQ-014 A norm and inner product:

$$\|x\|_2^2 = \langle x, x\rangle = \sum_{i=1}^{d} x_i^2 \tag{15}$$

Vector norms recur throughout the optimisation literature.

CORPUSMARK-EQ-015 A binomial coefficient expanded as a product:

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!} = \prod_{i=1}^{k} \frac{n-i+1}{i} \tag{16}$$

Combinatorial identities stress factorial and product notation.

CORPUSMARK-EQ-016 A nested radical with a closed form:

$$\sqrt{a+\sqrt{a+\sqrt{a+\cdots}}} = \frac{1+\sqrt{1+4a}}{2} \tag{17}$$

Nested radicals produce deeply stacked glyphs.

CORPUSMARK-EQBLOCK-001 A block of several short stacked equations whose bounding box is close to square, the shape that geometry misreads as a figure rather than mathematics:

$$\begin{aligned} Y_1 &= X_1 \\ Y_2 &= X_1 + X_2 \\ Y_3 &= X_1 + X_2 + X_3 \\ Y_4 &= X_1 + X_2 + X_3 + X_4 \end{aligned} \tag{18}$$

A cumulative construction, one term added per line.

CORPUSMARK-EQBOX-001 A display equation set inside a shaded definition box, the way a textbook frames a key formula. The shading makes the whole region a graphic, and the words must be kept:

Definition:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

with $df_{\text{between}} = p - 1$ and $df_{\text{within}} = N - p$.

The boxed form recurs wherever a definition is highlighted.

CORPUSMARK-INLINE-001 Inline mathematics stays in the text layer and must not become a crop: the identity $e^{i\pi} + 1 = 0$ and the bound $0 \leq p \leq 1$ both sit inside this running sentence, alongside symbols such as α, β and $\sum_i x_i$, none of which the extractor pulls out into an image.

CORPUSMARK-TBL-001 A ruled numeric table:

Condition	Mean	SD
Baseline	12.40	1.15
Treatment	15.82	1.640
Control	12.11	1.09

The table above reports descriptive statistics.

CORPUSMARK-FIG-001 A vector figure drawn from primitives, with a caption beneath it:

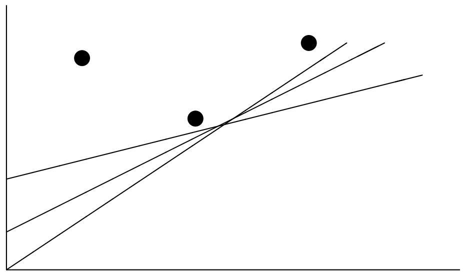

Fig. 1 Line segments and plotted points, captioned in the style a converted article uses.

CORPUSMARK-FIG-002 A bar chart of a discrete probability distribution, the commonest figure in a statistics text:

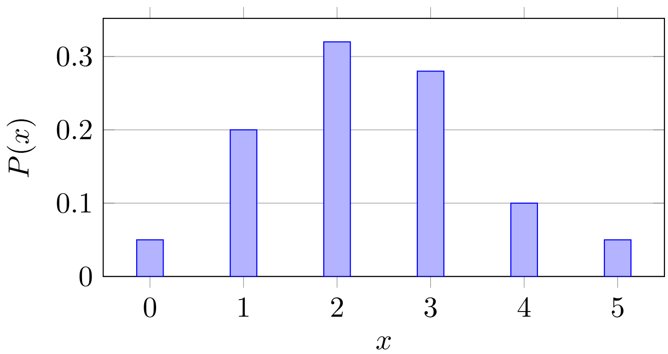

Fig. 2 A probability mass function.

CORPUSMARK-FIG-003 A normal density curve with two shaded tail regions, exactly the figure a text uses to introduce significance testing:

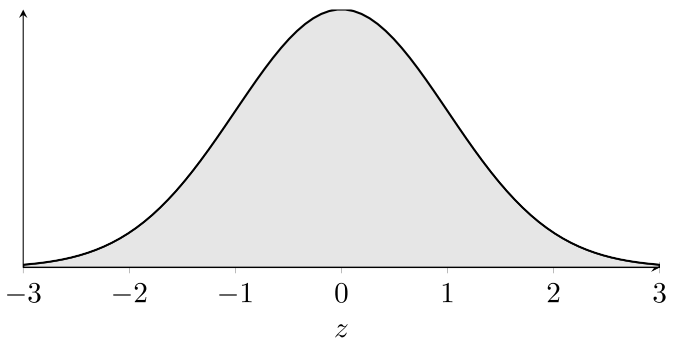

Fig. 3 The standard normal density.

CORPUSMARK-FIG-004 A line plot drawn on a full coordinate grid. The grid lines are the trap: a classifier can mistake the ruled background for a table.

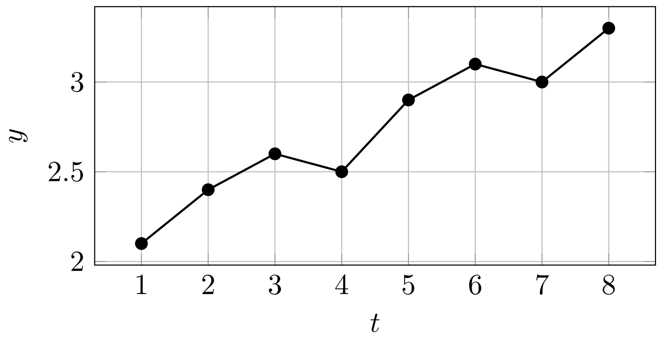

Fig. 4 A response curve on a grid.

CORPUSMARK-FIG-005 A Venn diagram of two overlapping events inside a sample space:

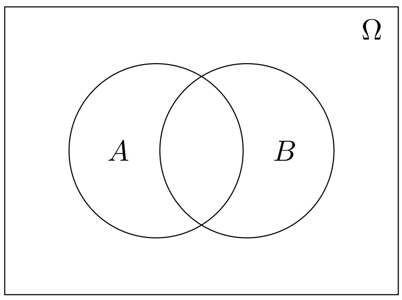

Fig. 5 Two events and their intersection.

CORPUSMARK-FIG-006 A path diagram with a latent variable and its indicators, the schematic a structural-equation-modelling paper prints:

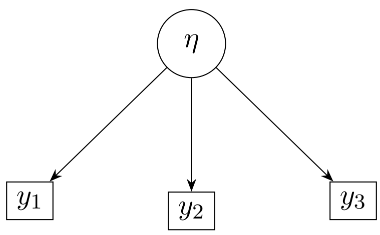

Fig. 6 A one-factor measurement model.

CORPUSMARK-TBL-002 A contingency table embedded as a raster image built by the corpus tool. A raster table has no text layer, so – unlike a LaTeX tabular – it is extracted as an image crop, reproducing the real failure mode where a publisher's table defeats the text extractor:

	Y = 0	Y = 1
X = 0	0.16	0.34
X = 1	0.24	0.26

Tab. 2 A joint probability table.

CORPUSMARK-TBL-003 A denser computational table of fractions, also embedded as a raster image. A wall of fractions like this is easily misread as a single large equation:

x	p(x)	F(x)
0	1/6	1/6
1	2/6	3/6
2	3/6	6/6

Tab. 3 A cumulative distribution, tabulated.

CORPUSMARK-NEG-001 A full-width decorative band follows. It is wide and short – the same shape as a display equation, and the reason a classifier keyed on aspect ratio alone fails – and it contains nothing to read:

That band carries no content.

CORPUSMARK-NEG-002 A narrow vertical spine bar follows, of the kind publishers print down a page edge:

That bar carries no content either.

CORPUSMARK-NEG-003 A solid square block, standing in for a logo or a masthead device:

Blocks like this are pure decoration.

CORPUSMARK-NEG-004 A separator band of two thick rules, wide and short like the first negative:

Nothing above this line needs recognising.

CORPUSMARK-DEC-001 A small empty framed box, the shape an extracted form-field or checkbox glyph takes – content-free, and small enough that a size gate should drop it before any model sees it:

An empty box carries nothing to recognise.

CORPUSMARK-DEC-002 A vertical gradient bar, the decorative device a publisher runs down a column. It is a graphic with no text, and must never be sent to an OCR model as a formula:

The bar is pure decoration.